Decomposition theorem

Eigen value Decomposition: In linear algebra eigen decomposition or sometimes spectral decomposition is the factorization of a matrix into a canonical form, whereby the matrix is represented in terms of its eigen values and eigen vectors. It can be written as

$A = PDP^{-1}$.

Example - 42: sol:

Given that,

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$$

The characteristic $= A - \lambda I$

$$= \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4-\lambda & 0 \\ 0 & 4-\lambda \end{bmatrix}$$

The characteristic polynomial,

$$|A - \lambda I| = \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix}$$

$$= (4-\lambda)(4-\lambda) - 1$$

$\therefore |A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (4-\lambda)(4-\lambda) - 1 = 0$$

$$\Rightarrow 16 - 4\lambda - 4\lambda + \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 15 = 0$$

$$\Rightarrow \lambda(\lambda - 5) - 3(\lambda - 5) = 0$$

$$\Rightarrow (\lambda - 5)(\lambda - 3) = 0$$

$$\lambda = 5, 3$$

$$D = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$$

Let a non-zero vector $v = \begin{bmatrix} x \\ y \end{bmatrix}$

we have

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x + y \\ x + 4y \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \end{pmatrix} \quad [\text{for } \lambda = 3]$$

$$\Rightarrow \begin{aligned} 4x + y &= 3x \\ x + 4y &= 3y \end{aligned}$$

$$\Rightarrow \begin{aligned} x + y &= 0 \\ x + y &= 0 \end{aligned}$$

Since 1st and 2nd equation are identicant so we can disregard one of them

$\therefore x + y = 0$

In echelon form, there are only one equation in too unknown then the System has a non-zero solution and in particular $(2-1) = 1$ free variable which is y

Let, $y = 1$ then $x = -1$

$\therefore$ Eigen vector $v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ for $\lambda = 3$

again for $\lambda = 5$

Let a non zero vector $v_2 = \begin{bmatrix} x \\ y \end{bmatrix}$

we have $Av_2 = \lambda v_2$

$$\Rightarrow \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 5 \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\Rightarrow 4x + y = 5x$$
$$x + 4y = 5y$$

$$\Rightarrow x - y = 0$$
$$x - y = 0$$

Since 1st and 2nd equation are identical so we cand disregard one of them.

$$\therefore x - y = 0$$

In echelon form there are only one equation in two unknown then the system has a non-zero solution and in particular $(2-1) = 1$ free variable which is y.

Let $y = 1$ then $x = 1$

$\therefore$ Eigen vector $V_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ for $\lambda = 5$

P be a matrix of eigen vectors of a given matrix A,

$$\therefore P = [V_1, V_2]$$

$$= \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$|P| = \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -1 - 1 = -2$$

Now the co-factor of given matrix P are given below:

$C_{11} = 1 \qquad C_{12} = -1$

$C_{21} = -1 \qquad C_{22} = -1$

So the co-factor of matrix P. $P^c = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$

The adjoint of matrix P.

$$Adj(P) = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

Here,
the inverse matrix $P^{-1} = \frac{Adj(P)}{|P|}$

$$= \frac{1}{-2} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Then A can be writtens as an eigen decomposition,

$$A = PDP^{-1}$$

$$= \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Justification

Here,

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$$

and $PDP^{-1} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$

$$= \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -\frac{3}{2} & 3/2 \\ 5/2 & 5/2 \end{bmatrix}$$

$$= \begin{bmatrix} 3/2 + 5/2 & -3/2 + 5/2 \\ -3/2 + \frac{5}{2} & 3/2 + 5/2 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} = A \text{ [Justified]}$$

Single value Decomposition:

$$A = USV^T$$

A = Given matrix
U = Matrix of eigen vectors of the matrix AA^T
V = Matrix of Eigen vectors of the Matrix AA^T or A^TA
V^T = Transpose of the matrix V

Example- 43

Sol:

Let,

$$A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$$

In order to find u, Now we need to find the eigen vectors for AA^T

Given,

$$A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$$

$$A^T = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

$$\text{So, } AA^T = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

$$= \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix} \quad \dots\dots (1)$$

$$\therefore AA^T - \lambda I = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{bmatrix}$$

So the characteristic equation

$$|AA^T - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (8-\lambda)(2-\lambda) = 0$$

$$\Rightarrow \lambda = 8, 2$$

The diagonalize matrix, $D = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}$

The matrix of singular values,

$$S = \sqrt{D}$$

$$= \begin{bmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \quad \cdots\cdots (11)$$

we have,

$$Au_1 = \lambda u_1$$

Let, $u_1 = \begin{bmatrix} x \\ y \end{bmatrix}$

$$\therefore AA^T u_1 = \lambda u_1$$

$$\Rightarrow \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 8 \begin{bmatrix} x \\ y \end{bmatrix} \quad [\text{when } \lambda = 8]$$

$$\Rightarrow 8x = 8x$$

$$2y = 8y$$

$\Rightarrow 8x = 8x$

$y = 0$

$\therefore y = 0$

Let, $x = -1$

$\therefore u_1 = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ It is a unit vector or normal vector since the length of u_1 is $|u_1| = \sqrt{(-1)^2 + (0)^2} = \sqrt{1} = 1$

again, $u_2 = \begin{bmatrix} x \\ y \end{bmatrix}$

Here,

$AA^T u_2 = \lambda u_2$

$\Rightarrow \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 2 \begin{bmatrix} x \\ y \end{bmatrix}$ [when $\lambda = 2$]

$\Rightarrow 8x = 2x$

$2y = 2y$

$\Rightarrow 6x = 0$

$2y = 2y$

$\therefore x = 0$

Let, $y = 1$

$u_2 = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ It is a unit vector or normal vector since the length of u_2 is $|u_2| = \sqrt{0^2+1^2} = \sqrt{2} = 1$

$$\therefore U = (u_1, u_2)$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \cdots (iii)$$

Now we need to find the eigen valaes of A^TA

$$\therefore A^TA = \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix} \cdots (iv)$$

$$\therefore A^TA = \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{bmatrix}$$

So the charactoristic equation $|A^TA - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (5-\lambda)(5-\lambda) - 9 = 0$$

$$\Rightarrow \lambda = 8, 2$$

$\therefore$ The diagonalize matrix,

$$D = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}$$

The matrix of singular values.

$$S = \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \cdots\cdots (v)$$

we have, $AV_1 = \lambda V_2$

Let,

$$V_1 = \begin{bmatrix} x \\ y \end{bmatrix}$$

$\therefore A^TA \; V_2 = \lambda . V_1$

$$\Rightarrow \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 8 \begin{bmatrix} x \\ y \end{bmatrix} \quad [\text{when } \lambda = 8]$$

$$\Rightarrow 5x - 3y = 8x$$

$$-3x + 5y = 8y$$

$$\Rightarrow -3x - 3y = 0$$

$$-3x - 3y = 0$$

$$\Rightarrow x + y = 0$$

$$x + y = 0$$

Since 1st and 2nd equation are identical so we can disregared one of them

$$x + y = 0$$

In echelon from there are only one equation in two unknown then the System has a ~~xxx~~ non-zero solution and in particular $(2-1)=1$ free variable, which is y

Let.

$y=1 \therefore x=-1$

$\therefore V_1 = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$; It is not unit vector or normal vector.

since the length of V_1 is $|V_1| = \sqrt{(-1)^2+1^2}$ $= \sqrt{2}$; this vector V_1 has to converted into unit vector by

we have.

$$V_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\therefore V_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

similerly

$$A V_2 = \lambda V_2$$

Let $V_2 = \begin{bmatrix} x \\ y \end{bmatrix}$

$$\therefore A^T A V_2 = \lambda V_2$$

$$\Rightarrow \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 2 \begin{bmatrix} x \\ y \end{bmatrix} \text{ [when } \lambda = 2$$

$\Rightarrow 5x - 3y = 2x$

$-3x + 5y = 2y$

$\Rightarrow x - y = 0$

$x - y = 0$

since 1st and 2nd equation are identical ; so we can disregard one of them

$\therefore x - y = 0$

In echelon form, there are only one equation for two unknown then the system has a non zero solution in particular $(2-1) = 1$

free - variable which is y.

Let $y = 1$ $\therefore x = 1$

$$\therefore V_2 = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$|V_2| = \sqrt{(1)^2 + (1)^2}$$

$$= \sqrt{2}$$

Thus,

$$V_2 = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

Now, we need to construct the argument orthogonal matrix V

$$V = \begin{pmatrix} v_1 & v_2 \end{pmatrix}$$

$$= \begin{pmatrix} -\sqrt{2}/\sqrt{2} & \sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{pmatrix} \text{ --- } (V_1)$$

$$\therefore V^T = \begin{pmatrix} -\sqrt{2}/2 & \sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{pmatrix}$$

From (II), (III) and (VI) Putting the values of U, S, V^T into the relation $A = USV^T$.

Therefore $A = USV^T$

$$A = \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} -\sqrt{2}/2 & \sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{pmatrix}$$

Justification test:

$$A = USV^T$$

$$= \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} -\sqrt{2}/2 & \sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2\sqrt{2} \times \sqrt{2}/2 + 0 & 2\sqrt{2} \times \sqrt{2}/2 + 0 \\ 0 + \sqrt{2} \times \sqrt{2}/2 & 0 + \sqrt{2} \times \sqrt{2}/2 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 2 \\ 1 & 1 \end{pmatrix}$$

LU decomposition:

Definition: If any linear System of equation can be performed as $Ax = b$ without raw inter changes then A can be factored into the product of a lower triangular matrix L and an upper triangular matrix U. $A = LU$ which is known as LU decomposition or LU factorization

Advantage of LU method

1. More economic if we need to solve many system with the same co-efficient matrix A but different right hand sides as we only need to evaluate L and U for one time. Once L and U are saved only the forward and backward substitution are needed to solve each system.

2) storage space may be economied. If A is not required after factorization we can store L and U in A.

Solve the linear system of using LU decomposition.

$$u_1 + 2u_2 + 3u_3 = 5$$
$$2u_1 - 4u_2 + 6u_3 = 18$$
$$3u_1 - 9u_2 - 3u_3 = 6$$

Sol:

Given that

The linear System

$$u_1 + 2u_2 + 3u_3 = 5$$
$$2u_1 - 4u_2 + 6u_3 = 18$$
$$3u_1 - 9u_2 - 3u_3 = 6$$

which can be written as

$$AU = b \quad \text{——} \quad (1)$$

where

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}, \quad u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} \text{ and}$$

$$b = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

Now According to LU decomposition

$$A = LU \text{ --- } ②$$

where,

$$L = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \text{ and } U = \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} l_{11} & l_{11} u_{12} & l_{11} u_{13} \\ l_{21} & l_{21} u_{12} + l_{22} & l_{21} u_{13} + l_{22} u_{23} \\ l_{31} & l_{31} u_{12} + l_{32} & l_{31} u_{13} + l_{32} u_{23} + l_{33} \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

$$\Rightarrow \therefore l_{11} = 1 \qquad l_{11} u_{12} = 2 \qquad l_{11} u_{13} = 3$$

$$\Rightarrow u_{12} = 2 \qquad \Rightarrow u_{13} = 3$$

$l_{31} = 3,\quad l_{31} u_{12} + l_{32} = -9$

$\Rightarrow l_{32} = -15$

$l_{31} u_{13} + l_{32} . u_{23} + l_{33} = -3$

$\Rightarrow l_{32} = -12$

Therefore,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -8 & 0 \\ 3 & -15 & -12 \end{pmatrix} \text{ and } u = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

From (i) & (ii)

we get,

$$LUu = b \quad \text{—— (iii)}$$

Let

$$Uu = V \quad \text{—— (iv)}$$

where

$$V = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

Now

from (3) and (iv)

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 2 & -8 & 0 \\ 3 & -15 & -12 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

$\rightarrow v_1 = 5$

$\therefore 2v_1 - 8v_2 = 18$

$3v_1 - 15v_2 - 12v_3 = 6$

$v_1 = 5$

and

$$2v_1 - 8v_2 = 18$$

$$\Rightarrow 2\times5 - 8v_2 = 18$$

$$v_2 = -1$$

again

$$3v_1 - 15v_2 - 12v_3 = 6$$

$$\rightarrow 3\times5 - 15(-1) - 12v_3 = 6$$

$$v_3 = 2$$

Now from (iv)

$$\Rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow u_1 + 2u_2 + 3u_2 = 5$$

$$u_2 = -1$$

$$u_3 = 2$$

$$\therefore u_3 = 2$$

$$u_2 = -1$$

And

$$u_1 + 2u_2 + 3u_2 = 5$$

$$\Rightarrow u_1 + 2(-1) + 3\times2 = 5$$

$$u_1 = 1$$

$$(u_1, u_2, u_3) = (1, -1, 2)$$

Vector and Scalars

Definition:

(i) vector : A vector is a quantify having both magnitude and direction. such as displacement velocity, force and acceleration etc.

(ii) Scalar: A scalar is a quantify having magnitude but no direction such as mass length ; time temparature and any real number.

(iii) unit vector: A unite vector is a vector having unite magnitude. If A is a vector with magnitude $A \neq 0$ then $\bar{A}/A$ is a unit vector having the same direction as A

$$\text{unit vector } \hat{a} = \frac{\bar{A}}{|\bar{A}|}$$

(iv) Dot product or (v) scalar product:

The Dot or Scalar product of two vectors $\bar{A}$ and $\bar{B}$ denoted by $\bar{A}\cdot\bar{B}$ is defined as the product of magnitude of $\bar{A}$ and $\bar{B}$ and the cosine of the angle θ between them

In symbals $A \cdot B = AB\cos\theta$. $0 \leq \theta \leq \pi$

(vi) vector field: If to each point (x, y, z) of a region R in space there corresponds a vector $\dot{v}(x, y, z)$ then v is called a vector function and we say that a vector field v has been defined in R

(vii) Scalar field: If to each point (x, y, z) of a region R in space there corresponds a number or scalar $\phi(x, y, z)$ then ϕ is called a scalar function of position or scalar point function and we say that a ~~sa~~ scalar field ϕ has been defined in R.

Example-22

a) Given that

$$r_1 = 3i - 2j + K$$

$$r_2 = 2i - 4j - 3K$$

$$r_3 = -i + 2j + 2K$$

the magnitude of $r_3 = |r_3|$

$$= |-i + 2\dot{j} + 2\hat{k}|$$

$$= \sqrt{(-1)^2 + 2^2 + (2)^2}$$